
Clustering Lead White References Across Graeco-Roman Genres: Visual Network Analysis from a TheSu XML Dataset

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Abstract

This paper addresses how the ideas conveyed by heterogeneous historical sources can be modelled as discourse networks for comparative visual network analysis. It presents a work-in-progress dataset of references to lead white (*ψιμθηται* / *cerussa*) in Graeco-Roman sources as a test case for a visualisation design developed for such discourse-network analyses. Lead white is a white pigment produced from lead, historically used for paints, as a cosmetic, and in medicine. The presentation builds on a visualisation and network-modelling methodology introduced at HNR 2024 (Morrone 2024), now applied to a richer dataset. The first version of the dataset (Morrone 2026), and the design of its network graphs, have recently been presented in a journal article (Morrone forthcoming). The dataset draws on passages from philosophical, technical, and literary works by Plato, Dioscorides, Plutarch, and others, selected for their descriptions of lead white, arguments about its properties, and recipes featuring lead white as an ingredient or end product (introduction to ancient sources about this white pigment and collection of recipes: Gliozzo and Ionescu 2022).

The dataset uses the *TheSu* XML encoding (Morrone 2023; <https://thesu.io>) to model descriptive statements ('theses') and procedural instructions ('sequences') conveyed by the passages alongside their discursive contexts ('supports' such as explanations, arguments, examples). Connections between these elements form network structures that can be visualised to analyse how lead white is presented, discussed, and referenced across different sources. When similar ideas appear in different passages or sources, the corresponding 'theses' are connected through an abstract 'proposition' node. This 'proposition' represents the idea that those statements attested in sources express in similar ways. Each recipe is modelled as a 'thesis' containing a 'sequence', which breaks the procedure down into constituent 'phases'. When recipes for the same product recur in different passages or sources with variants, their similarities and differences can be mapped step by step through correspondences between the 'phase' nodes of their respective 'sequences' within 'propositions'.

The visualisations, inspired by state-of-the-art argumentation mapping designs (Stede and Schneider 2019, 36–43), are optimised for interactive use in network analysis environments such as Gephi and for automatic clustering. Force-directed layouts make it possible to see at a glance, by proximity and through their connections, which sources may convey similar ideas in similar ways – potentially because they belong to the same tradition – and which stand apart. These visual impressions must be verified through close reading. The lead white corpus is currently too small for broader historical conclusions, but already provides

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some insights. One example concerns Plutarch, a Platonist author. Graph visualisations show that his discussion of lead white does not cluster with Plato's use of the substance as an example of whiteness. Instead, it connects with Dioscorides's medical and technical discussion. A second example concerns comparative recipe graphs: steps unmatched in other recipes are pushed towards the margins, and this makes peculiarities immediately visible. At this stage, the dataset primarily tests the value of a proposed visualisation design for comparative visual network analysis of heterogeneous historical sources. Larger corpora will provide farther-reaching conclusions.

Bibliography:

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Keywords: lead white, ancient Graeco, Roman sources, recipes and technical literature, discourse networks, TheSu XML, argumentation mapping, visual network analysis, force, directed layouts, digital humanities, history of science

Plato, *Lysis*,
217b-e

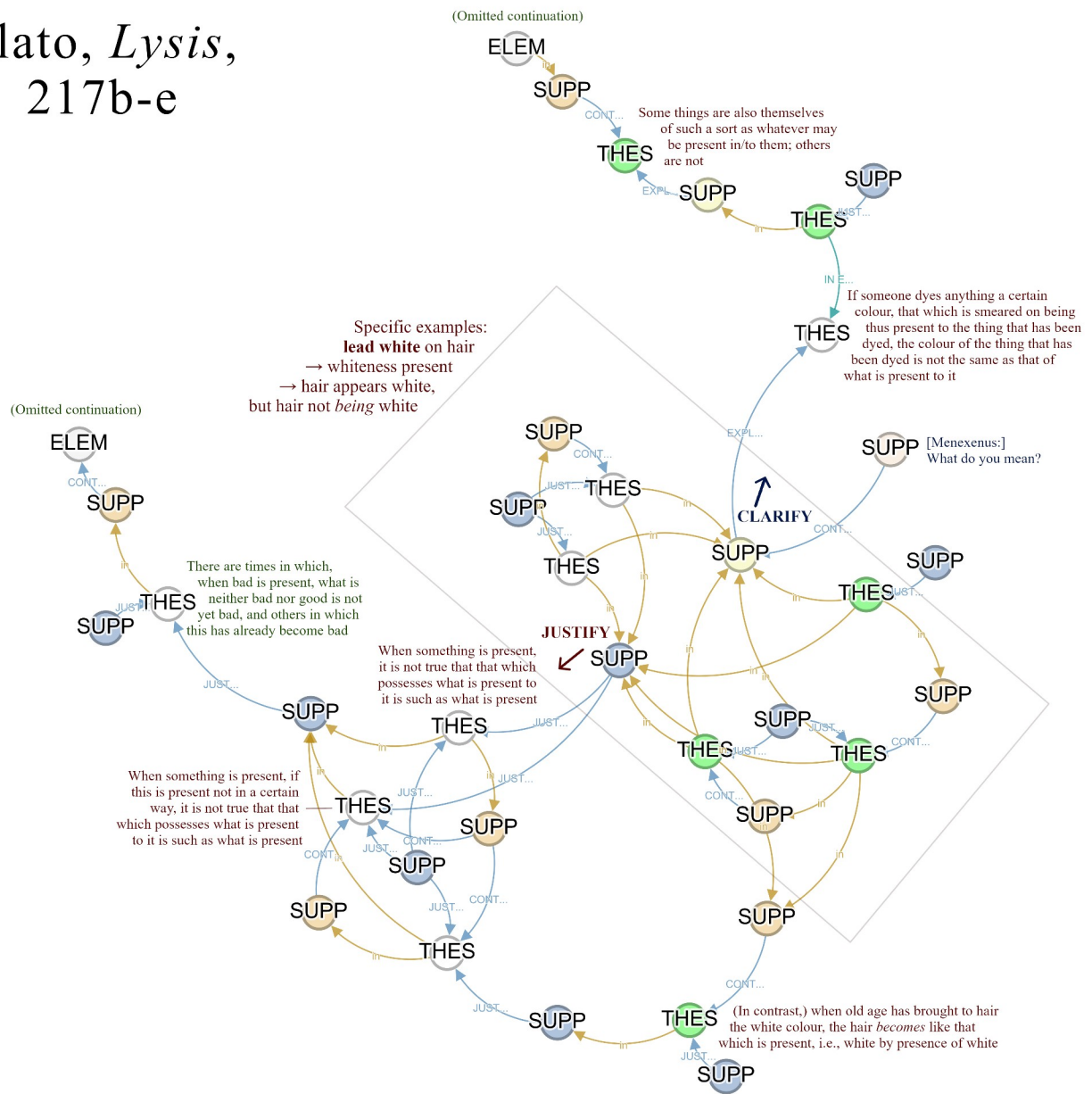


Figure 1. Single-source discourse network generated from a TheSu XML annotation of Plato, *Lysis* 217b–e. Nodes represent statements (‘theses’) and their discursive contexts (‘supports’), showing how Socrates’ example of lead white applied to hair first clarifies the relation between a pigment and a dyed object’s colour and then supports a broader argument. Generated in Gephi using the ForceAtlas2 layout. Captions, box, and arrows were manually added.

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PROPOSITIONS

- 1 Theophrastus's recipe for lead white (correspondences with)
- 2 Dioscorides's first recipe for lead white (correspondences with)
- 3 "Lead white is a harmful substance"
- 4 "Lead white is a cooling substance"
- 5 "Lead white is white in colour"

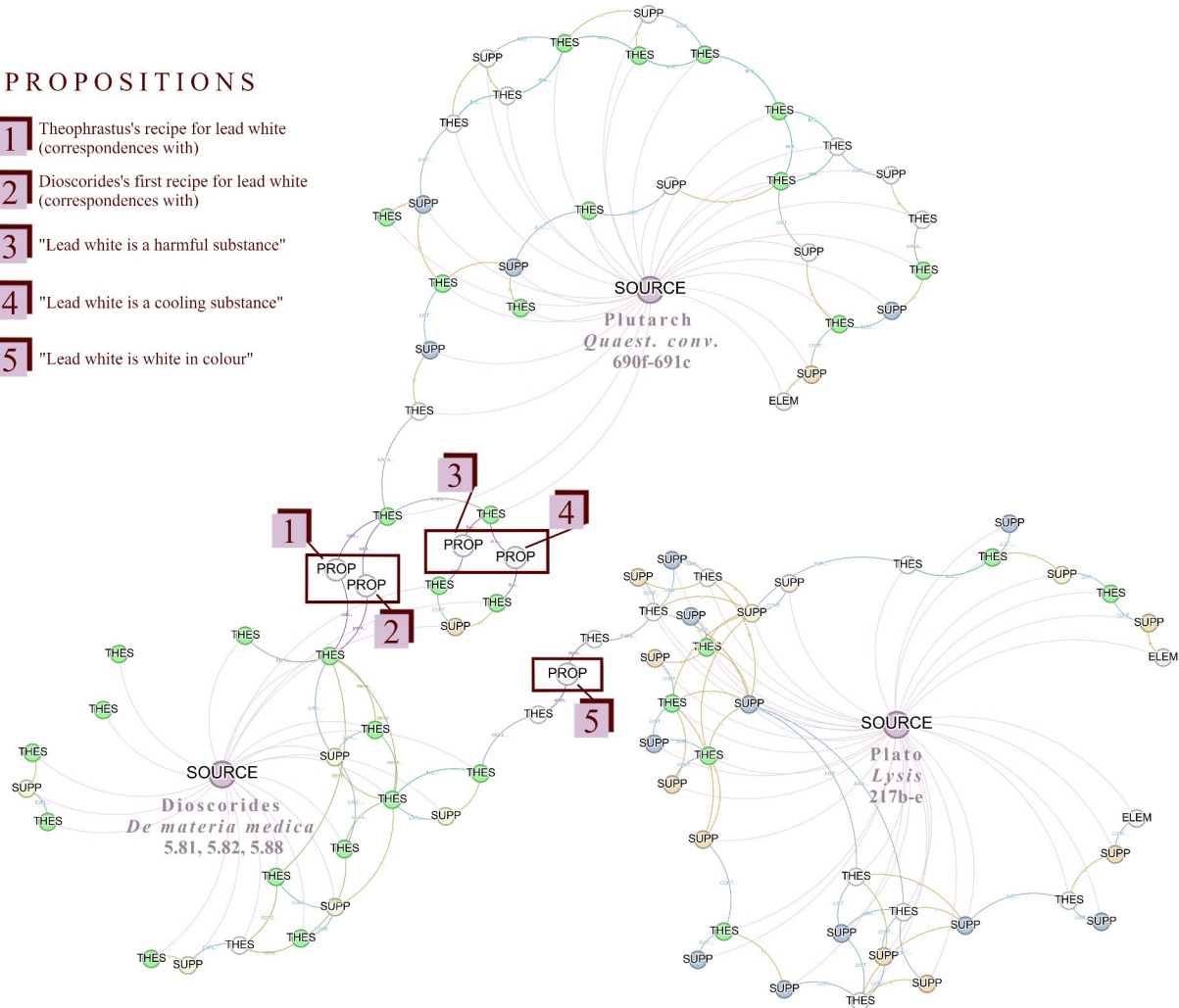


Figure 2. Comparative discourse network generated from a TheSu XML annotation of passages in Plato, Plutarch, and Dioscorides. Abstract 'proposition' nodes connect statements attested in the sources with similar content. Plutarch's chemical and pharmacological discussion links with Dioscorides rather than with Plato, even though Plutarch was a Platonist philosopher. Plato's chromatic example also separately links with Dioscorides. Generated in Gephi using the ForceAtlas2 layout. Captions and box

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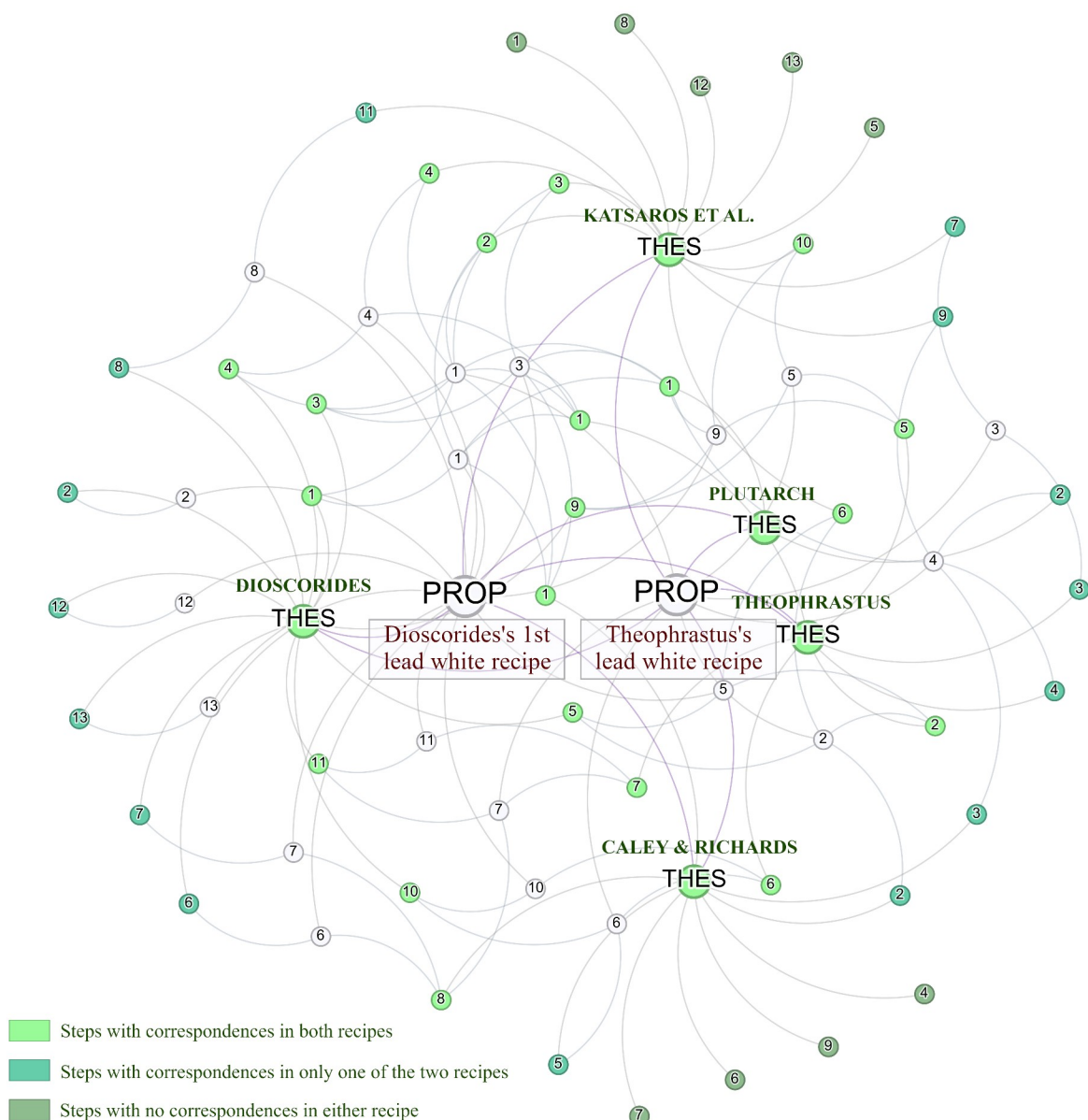


Figure 3. Comparative network of recipes for lead white generated from a TheSu XML annotation of Theophrastus, Dioscorides, Plutarch, and descriptions of modern replications by Caley and Richards (1956), and Katsaros, Liritzis, and Laskaris (2010). Green nodes represent procedures attested in the sources and their steps; purple nodes represent model procedures derived from Theophrastus and Dioscorides. In this force-directed layout, corresponding steps tend to appear closer together, while unmat